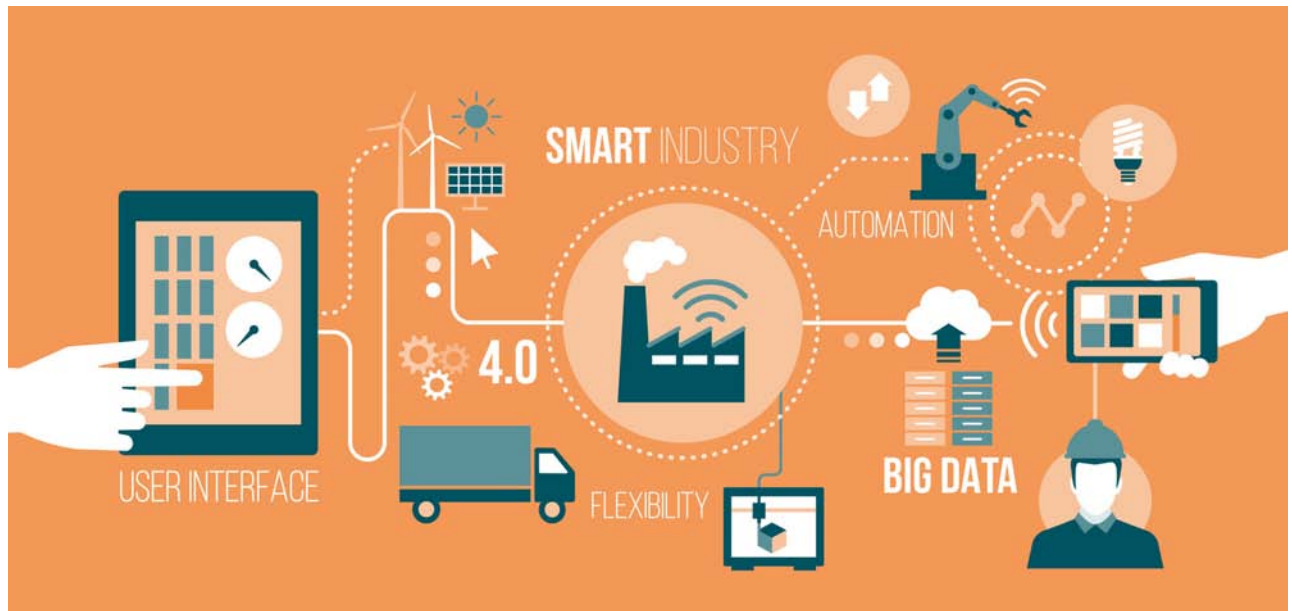


IoT Analytics: Making Sense of Big Data

This article covers the characteristics of IoT analytics and the challenges faced when adopting it in the industry. It also presents real world use cases of IoT analytics.



The standard definitions of IoT are:

- “A network of physical objects or things accessed through the internet, as defined by technology analysts and visionaries. These things contain embedded technology to interact with internal states or the external environment. In other words, when things can sense and communicate, it changes how and where the decisions are being made and who makes them.” – *Cisco*
- “The IoT can be viewed as a global infrastructure for the information society, enabling advanced services by interconnecting (physical and virtual) things based on existing and evolving interoperable information and communication technologies.” – *ITU*

- “An IoT system is a network of networks where, typically, a massive number of objects, things, sensors or devices are connected through communications and information infrastructure to provide value-added services via intelligent data processing and management for different applications (e.g., smart cities, smart health, smart grid, smart home, smart transportation, and smart shopping).” — *IEEE Internet of Things Journal*

Overall, the Internet of Things (IoT) is a network of devices which can sense, accumulate, and transfer data over the internet without any human intervention. IoT and data remain intrinsically linked together. Data consumed and produced keeps growing at an ever-expanding rate.

Internet of Things (IoT) analytics assesses the wide range of data collected from IoT devices. It analyses vast quantities of data and produces useful information called Big Data. It addresses large, diverse, complex, longitudinal, and/or distributed data sets generated from instruments, sensors, internet transactions, emails, videos, click streams, and/or all other digital sources available. The processing and integration of this heterogeneously generated data is done by IoT analytics.

IoT analytics helps gain insights into the operations of an enterprise, boosts productivity, increases the quality and flexibility of operations, and minimises material consumption.